

Hypothesis Testing

ANSWER KEY



The logic of hypothesis testing

- 1 State the null and alternative hypotheses for testing whether a new fertilizer increases the mean crop yield μ , given a historical mean $\mu_0 = 50$ kg. Classify the test as one-tailed or two-tailed.

$H_0 : \mu = 50$. $H_a : \mu > 50$. This is a one-tailed (right-tailed) test, since the alternative specifies a direction.

- 2 Explain the difference between a one-tailed and a two-tailed hypothesis test, giving an example scenario for each.

A one-tailed test is used when the alternative hypothesis specifies a direction (e.g. testing whether a new drug **increases** recovery rate). A two-tailed test is used when the alternative simply states the parameter differs from the claimed value, without specifying direction (e.g. testing whether a machine's mean output has changed at all, up or down).

- 3 A test has $H_0 : \mu = 75$ vs $H_a : \mu \neq 75$, and the sample data leads to rejecting H_0 at the 5% significance level. State, without further calculation, whether this data would also lead to rejection at the 1% level, justifying your answer.

Not necessarily — a stricter significance level (1%) requires stronger evidence (a smaller p -value) to reject H_0 . If the p -value is very small (well below 0.01), rejection would still occur at 1%; but if the p -value lies between 0.01 and 0.05, H_0 would be rejected at 5% but not at 1%.

Hypothesis test for a population mean (known σ)

- 4 A machine fills bags with a claimed mean of 500g, with known population standard deviation $\sigma = 15$ g. A sample of $n = 36$ bags has mean $\bar{x} = 494$ g. Test at the 5% level whether the mean is less than 500g, using $z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$ and critical value -1.645 .

$z = \frac{494 - 500}{\frac{15}{\sqrt{36}}} = \frac{-6}{2.5} = -2.4$. Since $-2.4 < -1.645$, we reject H_0 . There is significant evidence at the 5% level that the mean fill weight is less than 500g.

- 5 A population has known standard deviation $\sigma = 8$. A sample of $n = 64$ has mean $\bar{x} = 103$, against a claimed mean $\mu_0 = 100$. Test at the 1% level whether the mean exceeds 100, using critical value

$z^* = 2.326$.
 $z = \frac{103 - 100}{\frac{8}{\sqrt{64}}} = \frac{3}{1} = 3$. Since $3 > 2.326$, we reject H_0 . There is significant evidence at the 1% level that the mean exceeds 100.

Two-tailed tests for a population mean

- 6 A standardized test historically has mean $\mu_0 = 100$ with $\sigma = 15$. A sample of $n = 25$ students taking a new prep course has mean $\bar{x} = 107$. Test at the 5% level (two-tailed) whether the course changes the mean score, using critical values ± 1.96 .

$$z = \frac{107 - 100}{15/\sqrt{25}} = \frac{7}{3} \approx 2.333$$
Since $|2.333| > 1.96$, we reject H_0 . There is significant evidence at the 5% level that the prep course changes the mean score.
- 7 Using $z \approx 2.33$ from the previous problem, find the two-tailed p -value, given $\Phi(2.33) \approx 0.9901$, and confirm the conclusion.
 $p\text{-value} = 2(1 - \Phi(2.33)) = 2(1 - 0.9901) = 2(0.0099) = 0.0198$. Since $0.0198 < 0.05$, we reject H_0 , confirming the earlier conclusion.
- 8 A sample of $n = 50$ has mean $\bar{x} = 48.2$, against a claimed mean $\mu_0 = 50$ with $\sigma = 6$. Test at the 5% level (two-tailed) whether the mean differs from 50, using critical values ± 1.96 .

$$z = \frac{48.2 - 50}{6/\sqrt{50}} = \frac{-1.8}{0.8485} \approx -2.121$$
Since $|-2.121| > 1.96$, we reject H_0 . There is significant evidence at the 5% level that the mean differs from 50.

Hypothesis test for a Poisson mean

- 9 A website historically crashes at a rate of $\lambda_0 = 4$ per month. After a software update, 1 crash is observed in a randomly selected month. Test at the 5% level whether the crash rate has decreased, using $P(X \leq 1) \approx 0.0916$ for $X \sim \text{Po}(4)$.
 $H_0 : \lambda = 4$. $H_a : \lambda < 4$. $p\text{-value} = P(X \leq 1) = 0.0916$. Since $0.0916 > 0.05$, we do not reject H_0 . There is insufficient evidence at the 5% level that the crash rate has decreased.
- 10 Explain why an exact Poisson probability, rather than a normal approximation, is used to test a hypothesis about a Poisson mean when λ_0 is small.
When λ_0 is small, the Poisson distribution is noticeably skewed and a normal approximation would be unreliable, so exact Poisson probabilities give a more accurate p -value.

Further Poisson mean tests

- 11 A call center historically receives calls at $\lambda_0 = 6$ per hour. During a promotional event, 12 calls are received in one hour. Test at the 5% level whether the call rate has increased, using $P(X \geq 12) \approx 0.0201$ for $X \sim \text{Po}(6)$.
 $H_0 : \lambda = 6$. $H_a : \lambda > 6$. $p\text{-value} = P(X \geq 12) = 0.0201$. Since $0.0201 < 0.05$, we reject H_0 . There is significant evidence at the 5% level that the call rate has increased during the promotional event.
- 12 A factory's defect rate is historically $\lambda_0 = 3$ per shift. During a trial of new equipment, 8 defects are observed in one shift. Test at the 5% level whether the defect rate has increased, using $P(X \geq 8) \approx 0.0119$ for $X \sim \text{Po}(3)$.
 $H_0 : \lambda = 3$. $H_a : \lambda > 3$. $p\text{-value} = P(X \geq 8) = 0.0119$. Since $0.0119 < 0.05$, we reject H_0 . There is significant evidence at the 5% level that the defect rate has increased.

Hypothesis test for a binomial proportion

- 13** A company claims 40% of its customers use its mobile app ($p_0 = 0.4$). In a sample of $n = 15$ customers, 10 report using the app. Test at the 5% level whether the true proportion is greater than 40%, using $P(X \geq 10) \approx 0.0338$ for $X \sim B(15, 0.4)$.

$H_0 : p = 0.4$. $H_a : p > 0.4$. $p\text{-value} = P(X \geq 10) = 0.0338$. Since $0.0338 < 0.05$, we reject H_0 . There is significant evidence at the 5% level that more than 40% of customers use the app.

- 14** Explain why the binomial hypothesis test uses the exact binomial distribution rather than a normal approximation when n is small.

For small n , the binomial distribution can be quite skewed (especially if p_0 is far from 0.5), so a normal approximation would be unreliable; exact binomial probabilities give a more accurate p -value.

Further binomial proportion tests

- 15** A coin is tested for fairness ($p_0 = 0.5$). In 20 tosses, 6 heads are observed. Test at the 5% level (two-tailed) whether the coin is biased, using $P(X \leq 6) \approx 0.0577$ for $X \sim B(20, 0.5)$ (doubled for a two-tailed test).

$H_0 : p = 0.5$. $H_a : p \neq 0.5$. Two-tailed $p\text{-value} = 2 \times P(X \leq 6) = 2(0.0577) = 0.1154$. Since $0.1154 > 0.05$, we do not reject H_0 . There is insufficient evidence at the 5% level that the coin is biased.

- 16** A politician claims 60% of voters support a new policy ($p_0 = 0.6$). In a sample of 12 voters, only 3 support it. Test at the 5% level whether support is less than 60%, using $P(X \leq 3) \approx 0.0153$ for $X \sim B(12, 0.6)$.

$H_0 : p = 0.6$. $H_a : p < 0.6$. $p\text{-value} = P(X \leq 3) = 0.0153$. Since $0.0153 < 0.05$, we reject H_0 . There is significant evidence at the 5% level that support is less than 60%.

Type I and Type II errors, and significance level

- 17** In the website crash-rate test, describe what a Type I error would mean in context, and what a Type II error would mean.

A Type I error would mean concluding the crash rate has decreased when it actually hasn't (rejecting a true H_0) — wrongly crediting the software update. A Type II error would mean failing to detect a real decrease in the crash rate (failing to reject a false H_0) — missing genuine improvement from the update.

- 18** Explain what is meant by the significance level of a hypothesis test, and state the probability of a Type I error in a test conducted at the 5% significance level.

The significance level α is the probability of rejecting a true null hypothesis (a Type I error), chosen in advance by the person conducting the test. At the 5% significance level, $\alpha = 0.05$, so there is a 5% probability of a Type I error.

- 19 A researcher conducts a test at $\alpha = 0.10$ instead of the more common $\alpha = 0.05$. Explain how this affects the probability of Type I and Type II errors, compared to using $\alpha = 0.05$.
- Using a larger $\alpha = 0.10$ increases the probability of a Type I error (the rejection region is larger, so a true H_0 is more likely to be rejected by chance) but decreases the probability of a Type II error (it becomes easier to reject H_0 , so a false H_0 is more likely to be correctly rejected) — there is a direct tradeoff between the two error types as α changes.

A well-designed word problem: testing a new manufacturing process

- 20 This question has four parts. A factory's current process produces components with mean breaking strength $\mu_0 = 250$ N and known $\sigma = 20$ N. A new process gives a sample of $n = 40$ components with mean $\bar{x} = 257$ N. (a) State H_0 and H_a for testing whether the new process increases mean breaking strength (one-tailed). (b) Compute the test statistic z . (c) Using critical value $z^* = 1.645$ (5% level), state the conclusion. (d) Quality control also monitors defects; the historical rate is $\lambda_0 = 5$ per batch, but a batch from the new process has only 1 defect. Using $P(X \leq 1) \approx 0.0404$ for $X \sim \text{Po}(5)$, test at the 5% level whether the new process has reduced the defect rate.
- (a) $H_0: \mu = 250$. $H_a: \mu > 250$. (b) $z = \frac{257 - 250}{20/\sqrt{40}} = \frac{7}{3.1623} \approx 2.214$. (c) Since $2.214 > 1.645$, we reject H_0 : significant evidence at the 5% level that the new process increases mean breaking strength. (d) $H_0: \lambda = 5$, $H_a: \lambda < 5$. $p\text{-value} = P(X \leq 1) = 0.0404 < 0.05$, so we reject H_0 : significant evidence at the 5% level that the new process has reduced the defect rate.